Answer Key - Economics - Undergraduate Level Assessment 2 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) Linearly increasing; B) Inverted U-shape; C) Horizontal; D) Upward sloping; E) Vertical
   Note: Matt's demand curve for bratwurst would be horizontal because a constant increase in total utility implies that he derives the same additional satisfaction from each unit consumed, indicating that he is willing to purchase any quantity at the same price, reflecting perfectly elastic demand.

2. **Answer:** B
   Options: A) High elasticity of demand and supply; B) Decline and instability; C) Lack of technological progress
   Note: The correct answer, "decline and instability," reflects the fundamental economic challenges faced by farms, where long-run problems often stem from persistent declines in profitability and productivity, while short-run issues are characterized by fluctuations and uncertainties in market conditions and input costs.

3. **Answer:** A
   Options: A) Rich and poor countries will experience the same growth rates regardless of savings rates; B) Poor countries will have fluctuating growth rates independent of savings; C) The level of technology determines growth rates, not savings; D) The level of income has no impact on savings; E) Savings have no correlation with growth rates in any country
   Note: The Harrod-Domar model, in conjunction with the Keynesian theory of savings, suggests that investment drives growth, and since poor countries may have lower savings yet can attract foreign investment or aid, they can achieve growth rates similar to those of rich countries, regardless of their domestic savings rates.

4. **Answer:** E
   Options: A) the anticipated inflation rate plus the nominal interest rate; B) the nominal interest rate multiplied by the inflation rate; C) what one sees when looking at bank literature; D) the anticipated inflation rate minus the nominal interest rate; E) the nominal interest rate minus anticipated inflation
   Note: The real interest rate represents the purchasing power of money over time, and it is calculated by adjusting the nominal interest rate to account for the erosion of value caused by anticipated inflation.

5. **Answer:** D
   Options: A) A decrease in both supply and demand and an ambiguous change in interest rates.; B) An increase in supply lowering the interest rate.; C) A decrease in demand increasing the interest rate.; D) An increase in demand increasing the interest rate.; E) A decrease in supply increasing the interest rate.

Note: An increase in corporate optimism leads to greater business investment expectations, which increases the demand for loanable funds as firms seek loans to finance expansion, consequently driving up interest rates.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: Stagflation occurs when a leftward shift in the aggregate supply curve, often due to increased production costs or supply shocks, results in higher prices (inflation) while simultaneously causing a decrease in output (stagnation).

7. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the increase in real GDP per capita from $20,000to21{,}000$ represents a 5 percent increase calculated by the formula $(newvalue - oldvalue)/oldvalue \times 100$, specifically $(21,000 - 20,000)/20,000 \times 100 = 5\%$.

8. **Answer:** False
   Options: A) True; B) False
   Note: In the long run under perfect competition, price equals marginal cost and average total cost (P = MC = ATC), meaning the statement is false because it incorrectly asserts that price is greater than marginal cost and less than average total cost.

9. **Answer:** True
   Options: A) True; B) False
   Note: For an autoregressive process to be stationary, the roots of the characteristic equation must lie outside the unit circle to ensure that the impact of shocks to the system diminishes over time rather than amplifying, which would lead to non-stationarity.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because a negative externality occurs when the actions of individuals or firms impose costs on others, leading to a decrease in overall societal welfare rather than an increase.

Long Form Response

11. **Answer:** In monopolistic competition, the condition P = MC (price equals marginal cost) indicates that firms are producing at a level where the price consumers are willing to pay for an additional unit is equal to the cost of producing that unit. This leads to efficient allocation of resources, as firms are neither overproducing nor underproducing relative to consumer demand. However, unlike perfect competition, firms in monopolistic competition have some market power, allowing them to set prices above marginal costs, which can result in reduced output and higher prices compared to a perfectly

competitive market. Consequently, while the P = MC condition promotes efficiency in theory, the presence of differentiated products and brand loyalty can lead to inefficiencies, as firms may not fully exploit economies of scale. Overall, this dynamic can limit consumer choice and lead to a deadweight loss in the market.
Note: correct because it accurately describes how the condition P = MC in monopolistic competition leads to an efficient allocation of resources while also acknowledging the presence of market power that allows firms to set prices above marginal cost.

12. **Answer:** A poll tax of 2*perheadexemplifiesaregressivetaxsystembecauseitimposesthesameflatfee*
incomeindividualscomparedtowealthierindividuals, whocaneasilyabsorbthecost.Forexample, forsc
a year, a 2*taxrepresentsa*0.02
Note:correctbecauseitclearlyillustrateshowaflatfeepolltaxdisproportionatelyaffectslow−
incomeindividualsbyrepresentingasignificantlyhigherpercentageoftheirincomecomparedtowealth

End of Answer Key